Project: Modeling Insurance Claims Frequency and Severity

LACTU2150 - Statistical Analysis of Insurance Data

Introduction

In this project, you will analyze an insurance portfolio of motor third party liability (MTPL) policies from a Belgian insurer, covering data from the year 1997. Each observation represents a unique policyholder, with risk factors recorded at the beginning of the policy period.

The goal is to model the frequency and severity of claims using Generalized Linear Models (GLM) and Generalized Additive Models (GAM) and to identify the most significant features based on model selection criteria. You will explore:

1. A **Log-Poisson model** for the claim frequency (number of claims filed).
2. A **Log-Gamma model** for the average claim severity (average amount per claim).

Data Description

The dataset called `mtpl_be.rda` located in the subfolder **data** contains 163,231 policyholders and the following variables:

Variable	Description
nclaims	The number of claims filed by the policyholder.
exp	The fraction of the year 1997 during which the policyholder was exposed to the risk.
amount	The total claim amount in euros
avg	The average amount per claim in euros (defined as the total claim amount divided by the number of claims). NA if there is no claim.
coverage	Type of coverage provided by the insurance policy: TPL, PO, or FO. TPL = only third-party liability, PO = partial omnium = TPL + limited material damage, FO = full omnium = TPL + comprehensive material damage.
fuel	Type of fuel of the vehicle: gasoline or diesel.
sex	Gender of the policyholder: male or female.
use	Main use of the vehicle: private or work.
fleet	The vehicle is part of a fleet: yes or no.
ageph	Age of the policyholder in years.
power	Horsepower of the vehicle in kilowatt.
agec	Age of the vehicle in years.
bm	Level in the former compulsory Belgian bonus-malus scale (0 to 22, where a higher level indicates a worse claim history).

Variable	Description
long	Longitude coordinate of the center of the municipality where the policyholder resides.
lat	Latitude coordinate of the center of the municipality where the policyholder resides.

Objectives

1. **Frequency Modeling**: Use a Poisson regression with a log-link to model `nclaims` (number of claims).
2. **Severity Modeling**: Use a Gamma regression with a log-link to model `avg` (average amount per claim).

Instructions

Part 1: GLM Analysis

1. **Data Preparation**
   - For categorical variables like `coverage`, `fuel`, `sex`, `use`, and `fleet`, ensure they are set as factors.
   - Bin continuous variables such as `ageph`, `agec`,`power`, and `bm` to simplify the GLM analysis.
2. **Exploratory Data Analysis**:
   - Conduct exploratory analysis to visualize the relative frequency of each feature (`ageph`, `coverage`, `fuel`, `sex`, `use`, `fleet`, `ageph`, `power`, `agec` and `bm`). Provide one plot for each feature.
3. **Frequency Model (Poisson)**
   - Fit a Poisson regression model with a log-link to predict `nclaims`, using exposure (`exp`) as an offset.
   - Exclude geographic variables (`long`, `lat`) for this analysis.
   - Select features based on:
     - **AIC**: Use AIC to assess model complexity and fit.
     - **ANOVA test**: Perform nested model comparisons to determine the significance of additional variables.
     - **Cross-validation**: Implement a k-fold cross-validation to evaluate model stability and predictive performance.
4. **Severity Model (Log-Gamma)**
   - Fit a Gamma regression model with a log-link to predict `avg`.
   - Exclude geographic variables in this part as well.
   - Perform feature selection using AIC, ANOVA, and cross-validation similar to the frequency model.
5. **Report Results**
   - Summarize the models, highlighting the most important features based on BIC, ANOVA tests, and cross-validation.
   - Include model coefficients and interpretation of the effect of each significant predictor.
   - Plot the resulting insurance premium as a function of `ageph` and `bm`.

Part 2: GAM Analysis with geographic smoothing

1. **Data Preparation**
   - Consider all continuous factors, including `long` and `lat` in the dataset for modeling spatial effects.
2. **Frequency Model with Smoothing**
   - Fit a Poisson-GAM model, using `long` and `lat` as smoothed covariates to capture geographic patterns.
   - Use smoothing functions for continuous variables such as `ageph`, `agec`, `power`, and `bm`.
   - Evaluate the model using:
     - **AIC**: Check if the GAM model improves over the GLM by comparing AIC values.
     - **ANOVA test**: Conduct tests for the significance of smoothed terms.
     - **Cross-validation**: Use k-fold cross-validation to assess predictive accuracy.

3. **Comparison of GLM and GAM**
   - Summarize the impact of considering smoothing terms on model accuracy and interpretation.
   - Plot the different smoothed functions: `ageph`, `agec`, `power`, `bm`, the spatial effect of`long` and `lat`.

Grading Overview

The project is graded out of 20 points. The main structure of the grading scheme is as follows:

- **Compliance with instructions & quality of the report: 1.5 points** (deliverables, structure of the report, reproducibility).
- **Part 1 — GLM: 12.5 points** (data preparation 2 · exploratory data analysis 2 · frequency GLM 4 · severity GLM 3.5 · synthesis, interpretation and premium 1).
- **Part 2 — GAM (frequency + geography): 6 points** (model specification 2.5 · evaluation and model selection 2 · plots and interpretation 1.5).

Note that the GAM part is evaluated **on the frequency model only**.

Submission Requirements

1. **R Markdown Report**: Submit an R Markdown (`.Rmd`) file containing your full analysis, including code, output, and interpretations. Please verify carefully that the file compiles without errors!
2. **HTML Output**: Render the R Markdown file to an HTML document to include all outputs in a single, viewable file.
3. **Text file** : Include group names in a separate file.
4. **Zipped File**: Place the `.Rmd` and `.html`files, together with a separate file `group.txt` containing the names of the two group members, in a single zipped folder.

Anonymization requirement: the report itself (`.Rmd` and `.html`) must be fully anonymous. Do not include your names, student IDs, or any personal information anywhere in the report (leave the `author:` field of the YAML header empty). Also make sure that no personal file paths or other identifying traces appear in the code or in the outputs. The names of the group members must only appear in the `group.txt` file.

Please send the zipped file on Moodle by **Tuesday, December 1, 2026 at 23:59**.

Peer Review

After the submission deadline, each group will be asked to review the anonymous project of another group. Detailed instructions will be provided on **December 2, 2026**. The peer review deadline is **Tuesday, December 15, 2026 at 23:59**.
